Introduction to IoT and Embedded Systems

Module 3 Assignment

Here are details for **two popular microcontrollers** from different manufacturers, based on their **official datasheets**:

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1. ATmega328P (by Atmel / Microchip)

- **Clock Frequency:** Up to **20 MHz**
- **Bitwidth of the Datapath: 8-bit**
- **Flash Memory Size: 32 KB** (with 0.5 KB used for the bootloader)
- **Number of Pins: 28 pins** (in PDIP package)
- **Analog-to-Digital Converter (ADC):**
  - Yes, contains an **ADC**
  - **10-bit resolution**
  - **6 channels** (in 28-pin package)
- **Datasheet Reference:** **https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P_Datasheet.pdf**

2. MSP430G2553 (by Texas Instruments)

- **Clock Frequency:** Up to **16 MHz**

- **Bitwidth of the Datapath: 16-bit**
- **Flash Memory Size: 16 KB**
- **Number of Pins: 20 pins** (in PDIP package)
- **Analog-to-Digital Converter (ADC):**
  - Yes, contains an **ADC**
  - **10-bit resolution**
  - **8 channels**
- **Datasheet Reference:** **https://www.ti.com/lit/ds/symlink/msp430g2553.pdf**

Here is a comparison of the **Arduino** and **Raspberry Pi** platforms in terms of operating systems:

Arduino Platform

- **Operating Systems:**
  Arduino microcontrollers **do not run traditional operating systems**. Instead, they execute single-threaded programs written in C/C++ and uploaded via the **Arduino IDE**.
- **Real-Time Operating Systems (RTOS):**
  Although not typical, some Arduino boards (like Arduino Due or Portenta H7) can run lightweight RTOSs such as:
  - **FreeRTOS** (Open Source)
  - **Mbed OS** (Open Source)
- These are used for more complex, multitasking applications.

- **Open Source Status:**
    - **Arduino firmware and libraries:** Open Source
    - **FreeRTOS and Mbed OS:** Open Source

Raspberry Pi Platform

- **Operating Systems:**
  Raspberry Pi is a single-board computer that supports full Linux distributions and other OSes. Commonly used OSes include:
    - **Raspberry Pi OS** (formerly Raspbian) – Open Source
    - **Ubuntu Server/Desktop** – Open Source
    - **Kali Linux** – Open Source
    - **RetroPie** – Open Source (built on Raspbian)
    - **Windows 10 IoT Core** – Not Open Source
    - **LibreELEC** (media center OS) – Open Source
- **Open Source Status:**
    - Most OS options for Raspberry Pi are **open source** and based on **Linux**.
    - **Windows 10 IoT Core** is a **closed-source** OS offered by Microsoft for embedded development.